10TH CLASS PHYSICS — NEW SYLLABUS (PECTAA)

Chapter 10 — Thermal Physics

Solved Exercise | New Book (PECTAA 2026)

Exercise 10 (Solved)

This exercise tests the whole chapter: thermal expansion of solids, liquids and gases, specific heat capacity, evaporation and its cooling effect, latent heat and superconductivity. Every question is answered the way a full-marks answer should be written in the board paper.

A. Multiple Choice Questions

10.1 When the temperature of a rod of copper is increased, its length:

(a) remains the same
(b) increases
(c) decreases
(d) becomes double

Answer: (b) increases — on heating, the particles vibrate with larger amplitude and move a little apart, so the rod lengthens ($\Delta L = \alpha L_0 \Delta T$).

10.2 The amount by which unit length of a material increases when the temperature is raised to 1 °C, is called the coefficient of:

(a) cubic expansion
(b) volume expansion
(c) linear expansion
(d) none of these

Answer: (c) linear expansion — $\alpha = \Delta L / (L_0 \Delta T)$, i.e. the change in unit length per 1 °C rise of temperature.

10.3 What does a liquid turn into once it is evaporated?

(a) Gas
(b) Water vapours
(c) Air
(d) Both (a) and (b)

Answer: (a) Gas — evaporation is the change of a liquid into a gas at its surface without boiling; the vapours formed are the gaseous state of that liquid.

10.4 Which property determines how much heat a solid can absorb before its temperature changes significantly?

(a) Density
(b) Specific heat capacity
(c) Colour
(d) Hardness

Answer: (b) Specific heat capacity — $c = Q / (m\Delta T)$. A large c means more heat is needed for the same rise of temperature.

10.5 High-temperature superconductors typically involve which type of material?

(a) Pure metals
(b) Organic compounds
(c) Ceramics
(d) Liquid crystals

Answer: (c) Ceramics — the book states that high-temperature superconducting ceramics work at higher temperatures, such as 135 K.

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B. Short Answer Questions

10.1 What factors influence the thermal expansion of solids?

Three things decide how much a solid expands.

(i) The rise in temperature: a bigger temperature change gives a bigger expansion, because ΔL is directly proportional to ΔT.

(ii) The original length (or volume) of the object: a long rod expands more than a short one for the same heating.

(iii) The nature of the material: every material has its own coefficient of linear expansion α, which depends on the strength of its bonds. Aluminium ($\alpha = 2.4 \times 10^{-5}\ K^{-1}$) expands much more than pyrex glass ($0.4 \times 10^{-5}\ K^{-1}$).

These three factors are combined in the relation $\Delta L = \alpha L_0 \Delta T$.

10.2 What is latent heat of fusion?

Latent heat of fusion is the heat energy required to change 1 kg of a solid into liquid at its melting point, without any change in its temperature.

When ice melts, heat is used to break the strong bonds between the solid particles, so the thermometer shows no rise in temperature until the whole ice has melted. For ice the latent heat of fusion is $3.36 \times 10^5\ J\ kg^{-1}$, i.e. 3.36×10^5 J of heat is needed to melt 1 kg of ice at 0 °C.

It is calculated by $Q = mL_f$, where m is the mass in kg and L_f is the latent heat of fusion.

10.3 How do different materials exhibit varying thermal expansion behaviours?

Every material has a different coefficient of expansion, because the forces holding its particles together are different. In a material with strong bonds the particles are held tightly, so the expansion is small; where the bonds are weaker, the particles move apart more easily, so the expansion is large.

That is why the same heating expands aluminium more than steel, and steel more than glass. The book gives $\alpha = 2.4 \times 10^{-5}\ K^{-1}$ for aluminium, $1.2 \times 10^{-5}\ K^{-1}$ for steel and $0.4 \times 10^{-5}\ K^{-1}$ for pyrex glass.

The state of matter also matters. Solids have the strongest intermolecular forces, liquids weaker and gases the weakest, so for the same rise in temperature gases expand the most and solids the least.

10.4 How does evaporation contribute to cooling processes in everyday life?

In evaporation the fastest (most energetic) molecules leave the surface of the liquid as vapour and carry the heat away with them. The molecules left behind have less energy, so the liquid becomes cooler. This is called cooling caused by evaporation.

Examples from daily life: when we sweat, the sweat evaporates and takes heat from our skin, so we feel cool. Water kept in an earthen pitcher (suhii matka) cools because water oozes out and evaporates from its surface. Wet clothes dry and cool the surroundings, a hot cup of tea cools faster when poured into a saucer, and a wet

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towel on the head gives relief on a hot day.

10.5 What is latent heat of vaporization, and how is it calculated?

Latent heat of vaporization is the heat energy needed to change 1 kg of a liquid into gas at its boiling point, without changing its temperature.

When a liquid boils, the heat supplied is used to break the bonds between the liquid molecules so that they escape as gas; the temperature stays constant until all the liquid has changed into vapour. For water $L_v = 2.26 \times 10^6$ J kg^{-1}, so 2.26×10^6 J of heat converts 1 kg of water at 100 °C into steam.

It is calculated by $Q = mL_v$, where m is the mass in kg and L_v is the latent heat of vaporization. For example, for 0.5 kg of water, $Q = 0.5 \times 2.26 \times 10^6 = 1.13 \times 10^6$ J.

C. Constructed Response Questions

10.1 Why do bridges need expansion joints but glass jars with metal lids do not, even though both experience temperature changes? Discuss in terms of the nature of thermal expansion.

A bridge is a very long piece of metal, so its expansion is large. The change in length is $\Delta L = \alpha L_0 \Delta T$, and here L_0 is about 100 m. Taking steel with $\alpha = 1.2 \times 10^{-5}$ K^{-1} and a summer rise of 40 °C, $\Delta L = 1.2 \times 10^{-5} \times 100 \times 40 = 0.048$ m, that is about 5 cm.

If the two ends of the bridge were fixed, this 5 cm would bend or crack the structure. So small gaps called expansion joints are left, which close up in summer and open again in winter. Railway tracks are given the same gaps to avoid sun kinks.

A jar lid is only a few centimetres wide, so its expansion is a small fraction of a millimetre — nothing can bend or break. Here the expansion is actually useful. The metal lid has a bigger coefficient of expansion than the glass (brass 1.9×10^{-5} K^{-1} against glass 0.9×10^{-5} K^{-1}), so on pouring warm water the lid expands more than the neck of the jar and becomes loose. No joint is needed because the expansion is wanted, not prevented.

10.2 If two liquids are heated under same conditions, but one of them heats up faster, what does this indicate about the specific heat capacity?

It indicates that the liquid which heats up faster has the smaller specific heat capacity.

From $Q = mc\Delta T$, the rise in temperature is $\Delta T = Q / (mc)$. If the two liquids get the same heat Q and have the same mass m, the liquid with the smaller c must show the bigger ΔT. In other words, it needs less heat to warm 1 kg of it by 1 °C, so it heats up and also cools down quickly.

For example, equal masses of water and cooking oil heated by the same burner: the oil becomes hot first because its specific heat capacity is much smaller than the 4200 J kg^{-1} K^{-1} of water. That is also why metals (low c) heat quickly and wood (high c) heats slowly.

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10.3 Why does a liquid evaporate faster when the air is dry?

Evaporation goes on only as long as the air above the liquid can take more vapour. In dry air the humidity is low, so the air is far from being saturated and it quickly absorbs the vapour that leaves the surface. The vapour is carried away by the moving air, the space above the liquid stays unsaturated, and more and more high-energy molecules escape. So evaporation becomes faster.

In damp, humid weather the air is already full of water vapour. It cannot take much more, so vapour collects above the liquid, the rate of evaporation falls and wet clothes dry very slowly. This is why the same wet floor dries quickly on a dry summer day but stays wet on a rainy day.

10.4 When a piece of ice melts, the temperature does not increase even though heat is supplied. Where does the energy go, and why is this important in phase change?

The energy goes into breaking the bonds between the ice molecules, not into increasing their kinetic energy. In ice the molecules are held in fixed positions by strong intermolecular forces. The supplied heat does work against these forces, so the molecules become free to slide over one another and the solid changes into liquid. Because the average kinetic energy of the molecules does not increase, the thermometer keeps showing 0 °C until the whole ice has melted. This hidden heat is the latent heat of fusion, $Q = mL_f$.

This is very important in a phase change. Because a large amount of heat (3.36×10^5 J for every kg) is absorbed with no rise in temperature, ice is a very effective coolant: it cools drinks, cold packs and ice bags for fever, and it protects crops from frost. The same idea explains why steam at 100 °C causes a worse burn than boiling water at 100 °C — the steam gives up its latent heat of vaporization as it condenses on the skin.

10.5 Why do electric transmission wires sag in summer and tighten in winter?

The wires are made of metal, so they expand when heated and contract when cooled. In summer the Sun heats the long wire and its length increases by $\Delta L = \alpha L_0 \Delta T$. As the wire becomes longer while its two ends are fixed to the poles, the extra length hangs down as a sag.

In winter the temperature falls, ΔT becomes negative, the wire contracts and becomes shorter, so it pulls tight between the poles. If the wires were fixed without any allowance for this change, the tension in winter could break them; therefore they are strung with a small sag in the first place.

D. Comprehensive Questions

10.1 How does thermal expansion differ between solids, liquids, and gases? Provide examples to illustrate these differences.

All three states expand on heating because their particles gain kinetic energy and move apart, but the amount of expansion is very different. The table below shows the difference.

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Point	Solids	Liquids	Gases
Order of expansion	least	medium	greatest
Reason	particles held by very strong forces, only vibrate about fixed positions	forces are weaker, particles can slide but are still close	particles are far apart, forces are almost negligible
Expansion studied as	linear expansion ($\Delta L = \alpha L_0 \Delta T$) and volume expansion	real and apparent expansion in a container	volume expansion at constant pressure (Charles's law)
Example	a 100 m steel bridge grows by about 5 cm in summer, so expansion joints are left	mercury rises in the stem of a thermometer; liquid level rises in a flask	hot air balloon rises; a balloon expands when taken from a cold room into sunlight

Solids expand the least because their particles are tightly bound, so heating only increases the amplitude of their vibration. In liquids the bonds are weaker and the particles are freer, so the expansion is more. In gases the molecules are far apart with almost no binding force, so a gas expands a great deal for the same rise of temperature, and its volume is directly proportional to the absolute temperature at constant pressure.

10.2 What are some practical consequences of thermal expansion in everyday life, such as in construction materials or household appliances?

Thermal expansion is useful in some places and harmful in others. The main everyday consequences are:

1. Thermometers: the expansion of mercury or alcohol in a narrow tube is used to measure temperature.
2. Railway tracks: small gaps are left between the rails so that they can expand in summer; without the gaps the track bends and twists (sun kinks) and can cause derailment.
3. Bridges: expansion joints and rollers let the long steel girders expand and contract without cracking.
4. Metal lids on jars: warm water makes the metal lid expand more than the glass, so a tight lid opens easily.
5. Gas cylinders and aerosol cans: gas expands strongly on heating and the pressure rises, so these are kept away from heat and are fitted with pressure-relief valves to avoid explosion.
6. Pipes and plumbing: hot-water pipes can expand and burst at the joints, so loops and flexible joints are used; concrete roads are laid in slabs with gaps for the same reason.

Household appliances also show the effect: a thick glass tumbler may crack if boiling water is poured into it, because the inner surface expands first and the outer surface

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is still cold. Electric wires sag in summer and pull tight in winter for the same reason.

10.3 What is latent heat, and why is it important in phase transitions like melting and vaporization?

Latent heat is the heat energy required to change the state of a substance without changing its temperature. This heat is used to break or form the bonds between the particles instead of increasing their movement, so the thermometer shows no change while the state is changing.

For melting (solid to liquid) it is the latent heat of fusion, $Q = mL_f$; for vaporization (liquid to gas) it is the latent heat of vaporization, $Q = mL_v$. For water $L_f = 3.36 \times 10^5$ J kg^{-1} and $L_v = 2.26 \times 10^6$ J kg^{-1}.

Importance: (i) the temperature stays constant at the melting or boiling point, even though heat is being supplied, so ice keeps a drink at 0 °C until it has all melted; (ii) a large amount of heat is absorbed or released for a small mass, which is why steam carries so much energy in cooking and heating and why it burns worse than boiling water; (iii) refrigeration and sweating both work on this hidden heat, and (iv) the latent heat of vaporization is the reason a wet surface keeps a body cool.

10.4 Why do solids, liquids, and gases expand when heated? Explain with examples.

Heating increases the average kinetic energy of the particles of a substance. In a solid the particles are held in fixed positions, so they only vibrate with a larger amplitude about their mean positions and push their neighbours a little farther away — the solid expands. In a liquid the particles can slide over one another, so they move apart more freely and the expansion is bigger. In a gas the molecules are far apart and move at high speed, so they collide harder with the walls; the space they need increases and the gas expands. If the container is closed, the pressure rises instead.

Examples: a metal rod lengthens on heating (linear expansion), the mercury column rises in a thermometer, the air in a hot air balloon expands and becomes less dense so that the balloon rises, and a tight metal lid becomes loose when warm water is poured on it. A gas expands much more than a liquid, and a liquid more than a solid, for the same rise in temperature, because the intermolecular forces become weaker from solid to liquid to gas.

10.5 If we have an equal amount of water in a wide, shallow dish and a tall, narrow container kept at the same temperature, in which container will evaporation occur faster? Explain the reason scientifically.

Evaporation will occur faster in the wide, shallow dish.

Evaporation is a surface phenomenon: only those molecules that are at the surface of the liquid and have enough energy can escape into the air as vapour. In the shallow dish the same amount of water exposes a much larger surface area to the air, so many more molecules are at the surface and can escape at the same time. In the tall, narrow container the same water has a small surface, most of the molecules lie deep inside the liquid and cannot escape, so fewer molecules leave per second

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and evaporation is slow.
Since warm air also carries vapour away, the dish dries even faster. That is why we pour tea into a saucer to cool it, and why clothes are spread out instead of being left in a heap when they are to dry quickly.

E. Numerical Problems

10.1 A metal rod of length 1 m expands by 0.02 m when heated from 20 °C to 120 °C. Calculate its coefficient of linear expansion.

Given: $L_0 = 1$ m
$\Delta L = 0.02$ m
$\Delta T = 120 - 20 = 100$ °C $= 100$ K

To find: $\alpha = ?$

Formula: $\Delta L = \alpha L_0 \Delta T$

Solution: $\alpha = \Delta L / (L_0 \Delta T)$
$\alpha = 0.02 / (1 \times 100)$
$\alpha = 2 \times 10^{-4}\ K^{-1}$

Answer: $\alpha = 2 \times 10^{-4}\ K^{-1}$

The book prints the answer as $2.9 \times 10^{-4}\ K^{-1}$; with the given length, expansion and temperature change the value comes out $2.0 \times 10^{-4}\ K^{-1}$. In the paper, write the working first and then the answer you get.

10.2 A container holds 1 litre of water at 20 °C. What will be its volume at 80 °C, assuming water's coefficient of volume expansion is 2.1×10^{-4} per °C?

Given: $V_0 = 1$ litre $= 1000\ cm^3$
$T_1 = 20$ °C, $T_2 = 80$ °C
$\Delta T = 80 - 20 = 60$ °C
$\beta = 2.1 \times 10^{-4}\ °C^{-1}$

To find: $V = ?$

Formula: $\Delta V = \beta V_0 \Delta T$

Solution: $\Delta V = 2.1 \times 10^{-4} \times 1000 \times 60$
$\Delta V = 12.6\ cm^3$
$V = V_0 + \Delta V = 1000 + 12.6$
$V = 1012.6\ cm^3$

Answer: $V = 1012.6\ cm^3$ (1.0126 litre)

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10.3 A steel rod initially measures 2 m at 20 °C. If its coefficient of linear expansion is 1.2 × 10^{-5} °C^{-1}, what will be its length at 100 °C?

Given: $L_0 = 2$ m
$\alpha = 1.2 \times 10^{-5}$ °C^{-1}
$\Delta T = 100 - 20 = 80$ °C

To find: $L = ?$

Formula: $\Delta L = \alpha L_0 \Delta T$

Solution: $\Delta L = 1.2 \times 10^{-5} \times 2 \times 80$
$\Delta L = 1.92 \times 10^{-3}$ m = 0.00192 m
$L = L_0 + \Delta L = 2 + 0.00192$
$L = 2.00192$ m

Answer: $L = 2.002$ m

10.4 A steel bridge expands by 5 cm on a hot summer day. If the bridge originally spanned 100 m, what is the temperature change?

Given: $L_0 = 100$ m
$\Delta L = 5$ cm = 0.05 m
$\alpha = 1.2 \times 10^{-5}$ °C^{-1} (steel)

To find: $\Delta T = ?$

Formula: $\Delta L = \alpha L_0 \Delta T$

Solution: $\Delta T = \Delta L / (\alpha L_0)$
$\Delta T = 0.05 / (1.2 \times 10^{-5} \times 100)$
$\Delta T = 0.05 / (1.2 \times 10^{-3})$
$\Delta T = 41.67$ °C

Answer: $\Delta T = 41.7$ °C

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10.5 How much heat is required to raise the temperature of a 2 kg iron bar from 20 °C to 100 °C, given that the specific heat capacity of iron is 450 J kg^{-1} K^{-1}?

Given: $m = 2$ kg
$c = 450$ J kg^{-1} K^{-1}
$\Delta T = 100 - 20 = 80$ °C = 80 K

To find: $Q = ?$

Formula: $Q = mc\Delta T$

Solution: $Q = 2 \times 450 \times 80$
$Q = 72\,000$ J
$Q = 72$ kJ

Answer: $Q = 72$ kJ

10.6 How much heat is required to melt 500 g of ice at 0 °C into water at 0 °C? (Latent heat of fusion of ice = 3.36×10^5 J kg^{-1})

Given: m = 500 g = 0.5 kg
$L_f = 3.36 \times 10^5$ J kg^{-1}
Temperature stays at 0 °C

To find: Q = ?

Formula: $Q = mL_f$

Solution: $Q = 0.5 \times 3.36 \times 10^5$
$Q = 1.68 \times 10^5$ J
Q = 168 kJ

Answer: $Q = 1.68 \times 10^5$ J = 168 kJ

10.7 Calculate the heat required to completely vaporize 1 kg of water at 100 °C. (Latent heat of vaporization of water = 2.26×10^6 J kg^{-1})

Given: m = 1 kg
$L_v = 2.26 \times 10^6$ J kg^{-1}
Temperature stays at 100 °C

To find: Q = ?

Formula: $Q = mL_v$

Solution: $Q = 1 \times 2.26 \times 10^6$
$Q = 2.26 \times 10^6$ J
Q = 2.26 MJ

Answer: $Q = 2.26 \times 10^6$ J = 2.26 MJ

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Formulae to Remember

- Linear expansion: $\Delta L = \alpha L_0 \Delta T$, $L = L_0(1 + \alpha \Delta T)$
- Volume expansion: $\Delta V = \beta V_0 \Delta T$
- Heat gained or lost: $Q = mc\Delta T$
- Change of state: $Q = mL_f$ (melting), $Q = mL_v$ (boiling)